Patrapilli Vasanth Sai

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Summary

Full-Stack Developer with hands-on experience building web and mobile applications using React.js, React Native, Node.js, TypeScript, and PostgreSQL. Skilled in REST API development, real-time communication, database optimization, authentication, payment integration, and scalable application design. Experienced in developing reliable applications with Socket.IO, Prisma ORM, Docker, JWT, and role-based access control.

Technical Skills

Languages: Python, JavaScript, TypeScript

Frontend: React.js, React Native, HTML5, CSS3, Bootstrap

Backend: Node.js, Express.js, REST APIs, API Integration, JWT Authentication, RBAC

Database: PostgreSQL, MySQL, MongoDB, Prisma ORM

Real-Time: Socket.IO, Real-Time Event Communication, Firebase Cloud Messaging (FCM), Push Notifications

Algorithms & System Design: Driver Matching, Nearest-Driver Selection, Vehicle-Type Filtering, Idempotency, Pagination, Caching Concepts, Circuit Breaker, Retry Mechanisms, Timeout Handling, Fault Tolerance, Horizontal Scaling, Load Balancing.

DevOps / Tools: Docker, Docker Compose, Git, GitHub, VS Code, Android Studio

Experience

AICTE Virtual Internship Jan 2024 – Mar 2024
Frontend Developer Intern Remote

- Developed responsive UI screens using React.js, HTML5, CSS3, JavaScript, and Bootstrap with reusable compo-nents and responsive layouts.
- Integrated REST APIs and handled asynchronous responses, loading states, validation, and error scenarios.
- Performed API testing using Postman to validate request/response payloads, status codes, and endpoint func-tionality.
- Conducted manual frontend testing covering navigation, forms, responsiveness, UI functionality, and cross-browser compatibility.
- Collaborated with team members to resolve UI issues, improve usability, and deliver application features within development timelines.

Project

RideNow – Scalable Full-Stack Ride Booking Platform

Multi-application ride-booking platform with centralized backend architecture

Platform Architecture

- **Admin Dashboard Web:** Ride, driver, customer, pricing, payment, analytics, notification, staff, and audit management.
- **Customer Web:** Ride booking, vehicle selection, fare estimation, tracking, payments, notifications, offers, and ride history.
- **Rider / Driver Mobile:** Driver onboarding, profile/vehicle management, availability, ride requests, acceptance, ride lifecycle, notifications, and account controls.
- **Centralized Backend:** Node.js, Express.js, and TypeScript backend providing shared REST APIs, business logic, authentication, database, payments, notifications, and real-time services.

Core Engineering & Scalability

- Developed modular REST APIs with **JWT authentication, RBAC, validation, and centralized error handling**.
- Implemented **Socket.IO** for real-time driver availability, ride assignment, status updates, and event synchronization.
- Developed **driver-matching and nearest-driver selection** using vehicle-type filtering for Bike, Auto, and Cab.
- Implemented dynamic fare calculation with base fare, distance, time, and minimum-fare components.
- Optimized PostgreSQL using **database indexing, query optimization, Prisma ORM, pagination, and filtering**.
- Applied **idempotency, retry, timeout, and circuit-breaker concepts** for reliable ride and payment operations.
- Designed **stateless APIs, horizontal-scaling and load-balancing ready architecture** using Docker-based services.

Payments & Platform Operations

- Integrated **Razorpay** with UPI, card, and cash payment workflows, including payment-status tracking and refunds.
- Built admin controls for driver approvals, suspensions, customer management, pricing, payments, ratings, and audit logs.
- Implemented centralized **customer/driver notifications** with unread counts, targeted alerts, ride events, payment alerts, and offers.
- Added analytics for rides, revenue, driver earnings, completion/cancellation rates, and vehicle-category usage.

Infrastructure & Quality

- Containerized backend, PostgreSQL, Customer Web, and Admin Portal using **Docker and Docker Compose**.
- Performed **API testing, integration testing, database validation, build verification, and cross-application testing**.
- Used **Git/GitHub** for version control and project maintenance.

Technologies: React.js, React Native, TypeScript, JavaScript, Node.js, Express.js, PostgreSQL, Prisma ORM, Socket.IO, Firebase Cloud Messaging, REST APIs, JWT, RBAC, Razorpay, Docker, Docker Compose, Git, GitHub, Android Studio

Algorithms & System Design: Driver Matching, Nearest-Driver Selection, Vehicle-Type Filtering, Idempotency, Database Indexing, Query Optimization, Pagination, Circuit Breaker, Retry Mechanisms, Fault Tolerance, Horizontal Scaling, Load Balancing

Project Links: Live Project Demo | Project Technical Documentation
Source Code: View RideNow Source Code on GitHub

Certifications

- AICTE Virtual Internship – Frontend Development
- IBM Web Development Certification
- EduSkills Cyber Security Certification

Education

GITAM University — Aug 2022 – May 2024
Master of Computer Applications (MCA) — Visakhapatnam, Andhra Pradesh

Aditya Degree College — Jun 2019 – Apr 2022
B.Sc. Computer Science — Visakhapatnam, Andhra Pradesh